Low-Rank Spectral Fingerprints for AI-Generated Music Detection

Interim report 1: scoping memo

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Domain	Audio forensics / music provenance
Method family	STFT magnitude spectrograms; exact and randomized SVD
LA object	10-s mono clip; STFT magnitude $M \in \mathbb{R}_{\geq 0}^{1025\times 431}$; top $k = 20$ singular values
Algorithms	Exact NumPy/LAPACK SVD versus randomized SVD ($k = 20$, $p = 10$, one power step)
Data	Up to 500 paired clips from FakeMusicCaps and MusicCaps [1]

1 Problem and aim

The detection of AI-generated music has proven a problem for platforms, rights holders and listeners. In this work, we intend to distinguish AI-generated audio from human-generated audio. Given a short audio clip, we predict human-origin versus text-to-music-generated. We do so by checking whether the low-rank structure of its STFT magnitude spectrogram supports this distinction on held-out clips, rather than create a universal detector. Prior work also warns that strong controlled results may not generalize across manipulations or datasets [2, 3], so we try to balance classes and group each original/synthetic caption pair in the same data split.

We will sample up to 500 caption IDs from FakeMusicCaps and retrieve the matching human-made MusicCaps clips. We will choose one generated clip per ID, balance across the five generators where possible, and keep matched clips in the same split. If original clips are inaccessible, we will report the smaller paired sample rather than replace them with unmatched tracks. Success requires held-out balanced accuracy ≥ 0.65 with a 95% bootstrap confidence interval whose lower bound exceeds 0.50. The approximation must also satisfy $\|p^{\text{rand}} - p^{\text{exact}}\|_2 / \|p^{\text{exact}}\|_2 \leq 10^{-2}$ on at least 95% of clips; otherwise, randomized SVD will not be considered accurate enough for this feature pipeline.

2 Linear-algebra formulation

Each ten-second clip is converted to mono at 22.05 kHz. A centred short-time Fourier transform (STFT), using a 2048-sample window/FFT and hop 512, gives a magnitude matrix $M \in \mathbb{R}_{\geq 0}^{F\times T}$ with $F = 1025$ and $T \approx 431$. Its singular value decomposition is

$$M = U\Sigma V^\top, \qquad \sigma_1 \geq \sigma_2 \geq \cdots \geq 0.$$

We classify the first 20 normalized squared singular values, $p_i = \sigma_i^2 / \|M\|_F^2$, using logistic regression. The linear-algebra task is to compute or approximate this leading spectrum, which summarizes how spectrogram energy is distributed across orthogonal components. It may discard useful phase, local-frequency, and temporal cues, so classification is an empirical feasibility test.

3 Approaches and evaluation

Dense thin SVD (reference). NumPy/LAPACK gives a numerically accurate spectrum and is the reference for error measurement. For $F \geq T$, the cost is $O(FT^2 + T^3)$ flops and memory is $O(FT + T^2)$. Its advantage is reliability; its drawback is that it computes the full spectrum, and these moderate matrix sizes may leave little room for speedup.

Randomized SVD. Following Halko, Martinsson, and Tropp [4], we will implement range finding with target rank $k = 20$, oversampling $p = 10$ ($\ell = k+p = 30$), QR orthogonalization, and one power step ($q = 1$), then take the SVD of the reduced matrix. Its cost is approximately $O((2q+2)FT\ell + (F+T)\ell^2)$

flops and memory $O(FT + (F + T)\ell)$. It focuses work on the leading subspace, but can lose accuracy when singular values decay slowly or cluster; one power step may then be insufficient, and extra steps increase runtime. We will implement this randomized algorithm and the feature pipeline ourselves, while using NumPy / LAPACK for the dense reference and scikit-learn for logistic regression.

We will compare feature error and wall-clock time, then report balanced accuracy, ROC-AUC and bootstrap confidence intervals on a caption-grouped held-out set. A bandwidth-only classifier is a diagnostic baseline: if it performs well, labels may be explained by codec or spectral-bandwidth artifacts rather than music structure. Frobenius normalization reduces sensitivity to overall gain, but codec and bandwidth differences can still leak into labels; identical mono conversion, resampling and loudness normalization will be applied to both classes.

4 Risks and work plan

The main risks are unavailable matched originals, a flat singular-value spectrum, and leakage of generator or mastering artifacts into the classifier. If the target accuracy is missed, we will report this negative result rather than repeatedly tune on the test set. By 2 November, we will verify clip availability and lock the preprocessing and split protocol. By 16 November (interim report 2), we will implement the feature pipeline and randomized SVD, validate its error and runtime against dense SVD on a subset, and run an initial baseline. By 14 December, we will finish the paired evaluation (up to 500 IDs), compare against the bandwidth baseline, calculate confidence intervals, and discuss limitations in the final report. Iryna will lead data matching and preprocessing; Petro will lead randomized SVD and numerical validation; both will run the classification evaluation and write the report.

References

[1] L. Comanducci, P. Bestagini and S. Tubaro, "FakeMusicCaps: A Dataset for Detection and Attribution of Synthetic Music Generated via Text-to-Music Models," *Journal of Imaging*, vol. 11, no. 7, article 242, 2025. doi:10.3390/jimaging11070242.

[2] D. Afchar, G. Meseguer-Brocal and R. Hennequin, "AI-Generated Music Detection and its Challenges," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025. doi:10.1109/ICASSP49660.2025.10890655.

[3] Y. Li, Q. Sun, H. Li, L. Specia and B. W. Schuller, "Explainable detection of machine generated music and early systematic evaluation," *Scientific Reports*, vol. 16, article 13757, 2026. doi:10.1038/s41598-026-42133-7.

[4] N. Halko, P. G. Martinsson and J. A. Tropp, "Finding Structure with Randomness: Probabilistic Algorithms for Constructing Approximate Matrix Decompositions," *SIAM Review*, vol. 53, no. 2, pp. 217–288, 2011. doi:10.1137/090771806.